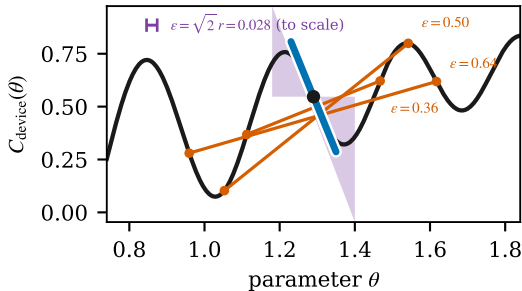


$H(\theta) = \theta Z_0 + X_0$ · Hamiltonian-level · $T/T_2^* = 0.5$
 best FD step $\epsilon^* = 0.11$; RMSE $\approx 28\%$ of $|\nabla C|$ · $r = 0.02$



— device landscape $C_{\text{device}}(\theta)$ $\text{FD setpoint cone at } \epsilon = \sqrt{2} r \text{ } (\pm 1\sigma = |\nabla C|)$
 — FD secants (wrong sign) — shift-rule tangent (PSR/NSR) $= \nabla C_{\text{device}}$